BTC/USDT Trading Pattern Analysis

Binance Spot, One-Second Kline Data

1 January 2023 to 5 November 2026, UTC

Research focus: Trading activity, volatility persistence and short-horizon return patterns

Yichen Wang

21Shares Take-Home Project

Project Workflow and Key Functions

Download — Retrieve and store daily data

Data Quality — Check integrity, coverage and values

Analysis — Identify patterns and generate charts

Validation & Tests — Evaluate robustness and verify code behaviour

Data Download and Storage

Retrieve BTC/USDT one-second candles through the public API.

Cache API responses to support interrupted downloads.

Store daily Parquet files with metadata and file hashes.



Data Quality Checks

Verify file integrity, timestamp continuity and OHLCV consistency.

Distinguish zero-volume observations from missing records.

Flag the documented halt as a warning and stop on unexpected errors.



Trading Pattern Analysis

Process daily files and calculate returns and time-based aggregates.

Examine intraday activity, regional sessions, weekends and return relationships.

Export summary tables and charts for interpretation.

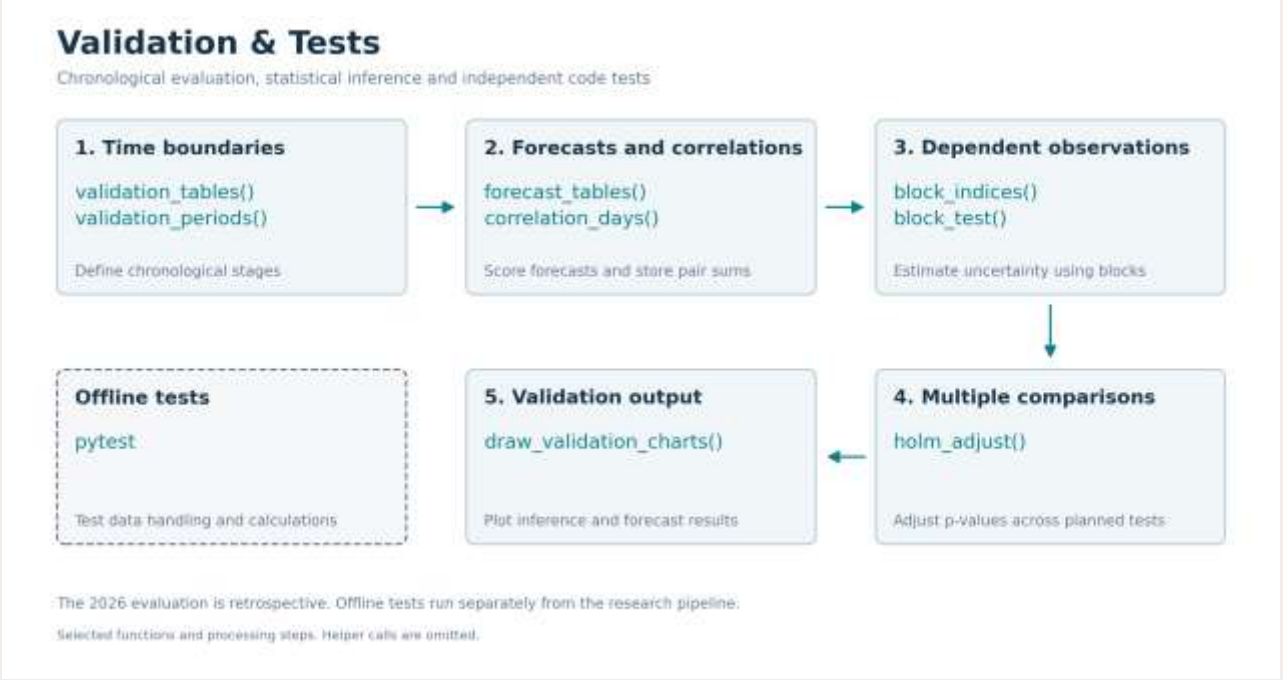


Research Validation and Code Tests

Fit models on 2023–2024, select on 2025 and evaluate on 2026.

Use block bootstrap and Holm adjustment to assess statistical evidence.

Run offline tests for downloads, data checks and analysis calculations.



Code Structure Summary

File	Responsibility	Key Design Choice
run.py	Configuration, CLI entry and task execution	Shared settings define the download and quality-check scope
download.py	API requests, pagination, caching and daily Parquet storage	Resumable downloads with controlled memory usage and request pacing
quality.py	File integrity, timestamp continuity and OHLCV checks	Explicit PASS, WARNING and FAIL outcomes
analysis.py	Aggregation, pattern research, validation and charts	Analysis starts after the data passes the quality gate

Data Pipeline

Daily processing: Process one day at a time to keep memory usage manageable.

Recovery and verification: Cache API responses to resume downloads and use file hashes to verify integrity.

Quality checks: Validate files, timestamps and OHLCV before analysis. Missing data unfilled.



Data Coverage and Quality

Metric	Result
Trusted daily partitions	1,405
Observed 1-second records	121,387,182
Observed coverage	99.996031%
Known missing seconds	4,818
Unexpected missing seconds	0
Complete analysis days	1,404
Complete comparison weeks	199

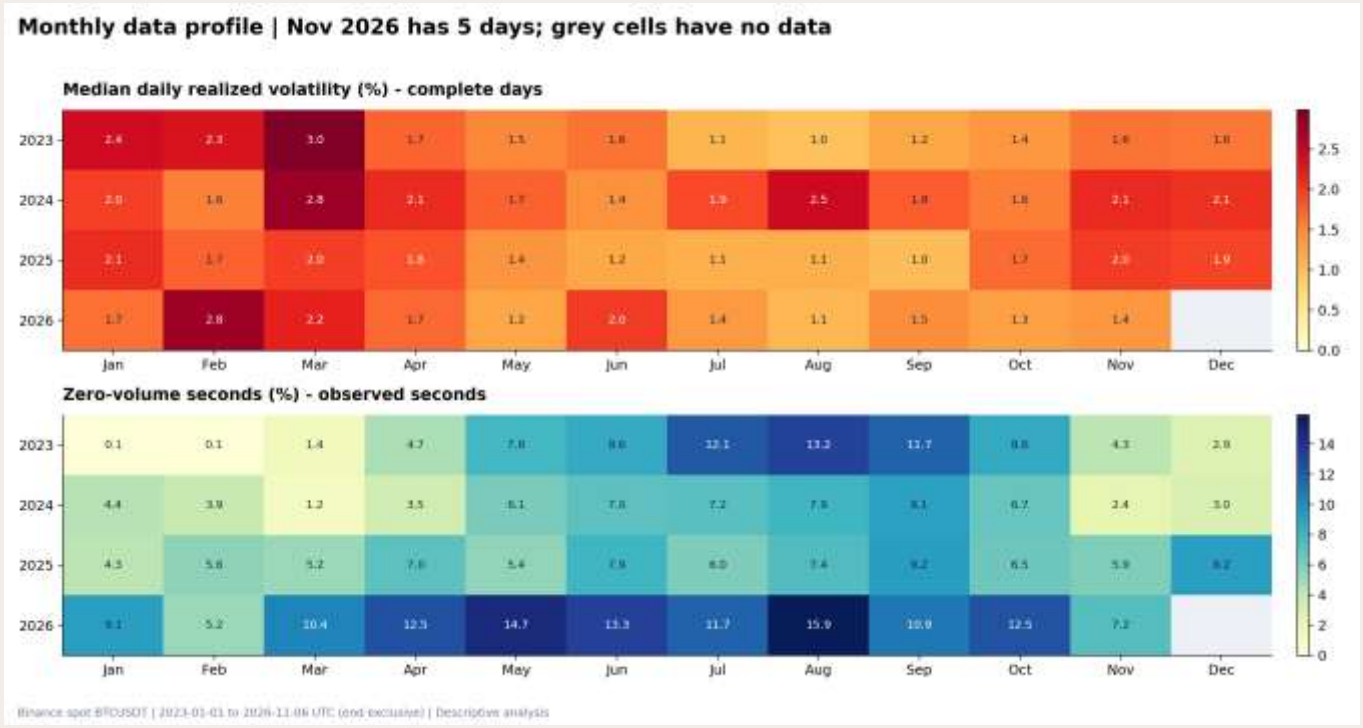
All missing seconds fall within one documented source gap on 24 March 2023.

Zero-volume seconds remain valid records. The known gap is marked as WARNING and left unfilled.

Returns never cross gaps. Seasonality uses complete UTC days; weekly comparisons use complete weeks.

Known gap: 24 March 2023, UTC
12:39:42 ≤ time < 14:00:00 · 80 min 18 sec

Activity and Volatility Over Time



Trading activity and volatility vary across months and years.

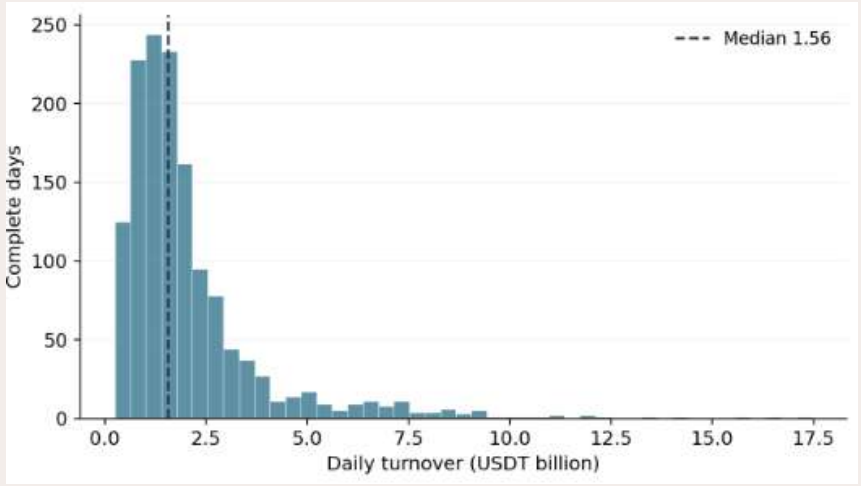
Zero-volume seconds were more common in 2026, while typical daily volatility rose much less.

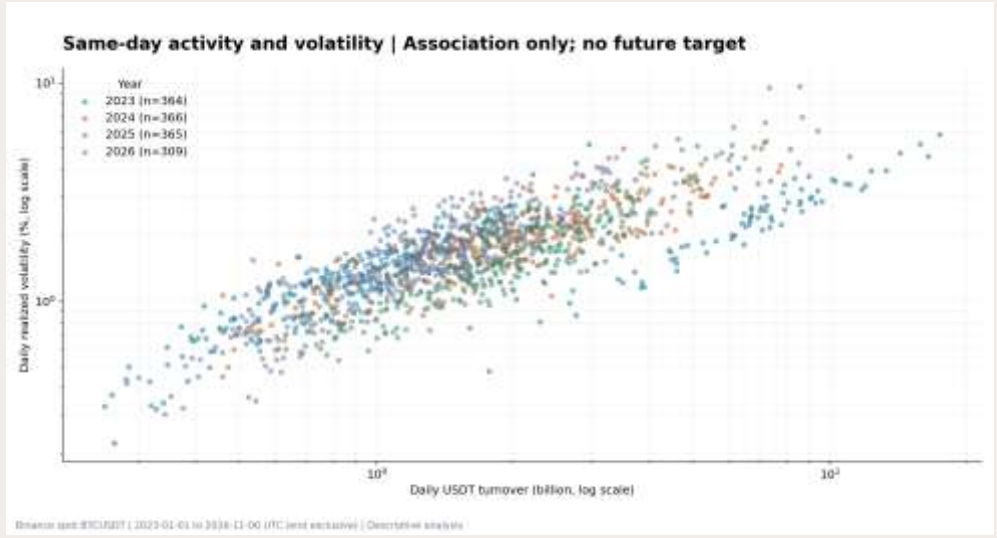
Daily Distributions and Activity–Volatility Association

Median daily turnover is **1.56 billion USDT**, compared with a mean of **2.09 billion**.

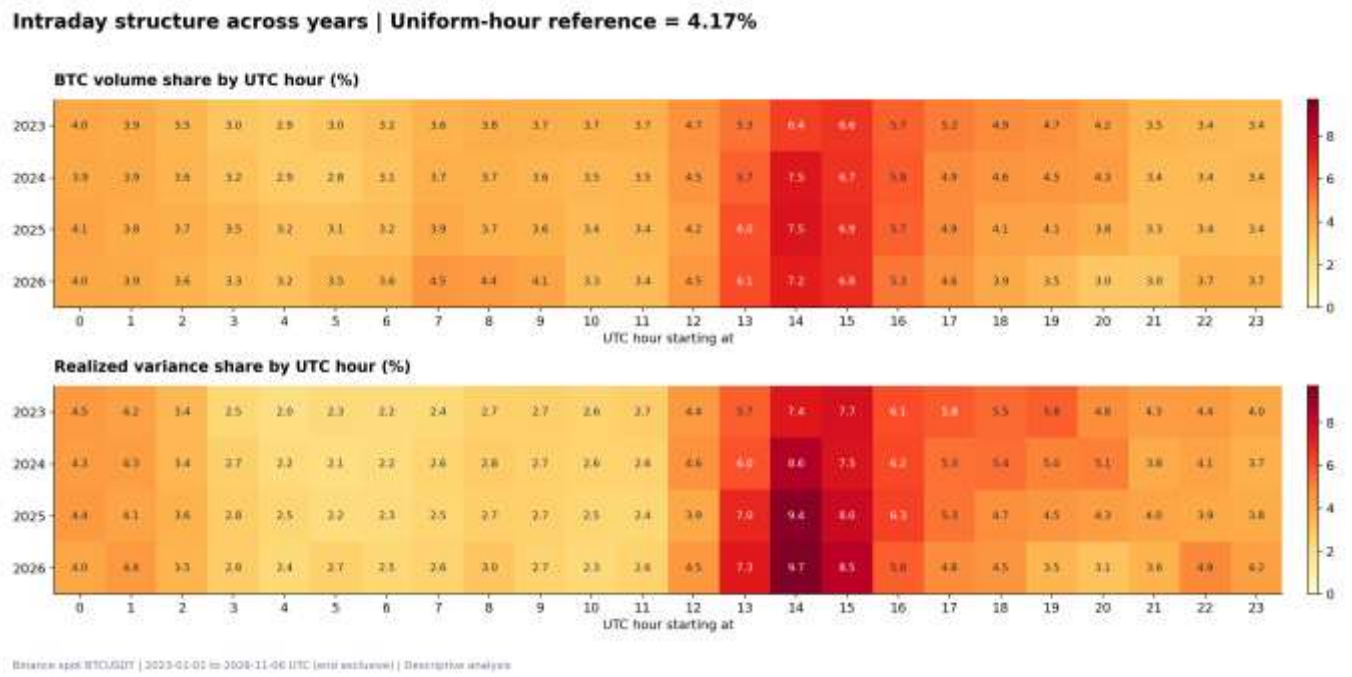
The distribution is **right-skewed**. A small number of highly active days lift the mean.

Higher-turnover days tend to have higher same-day realized volatility.





Trading Activity Concentrates in Specific UTC Hours



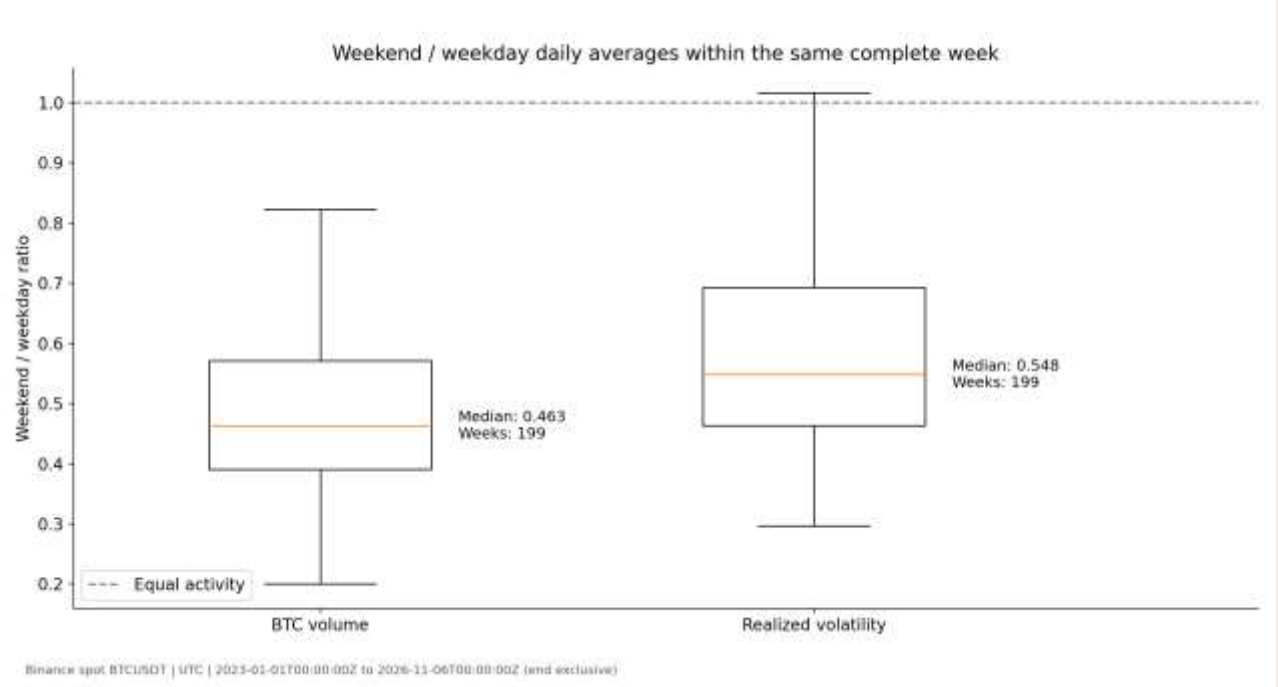
14:00–16:00 UTC represents **8.33%** of the day and accounts for **13.91%** of daily BTC volume and **16.66%** of daily realized variance.

Hourly shares are calculated within each complete UTC day, then averaged with equal weight.

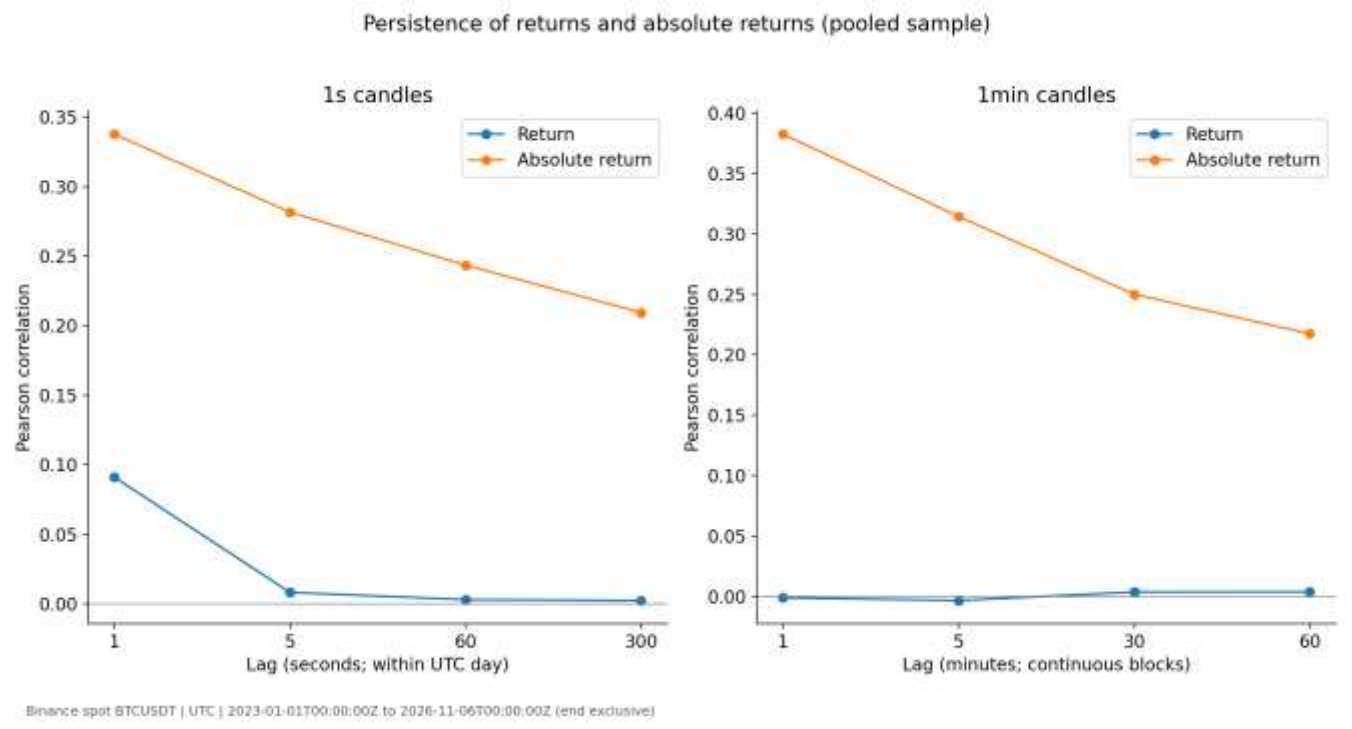
Activity and variance concentrate around the same UTC hours across years, with stronger concentration in variance.

Trading Activity and Volatility Are Lower on Weekends

- Within each of 199 complete weeks, compare the average daily level on weekends with weekdays.
- Median weekend/weekday ratios are 0.463 for BTC volume and 0.548 for daily realized volatility.
- Regional windows also show lower weekend activity. These data do not measure spreads or trading costs.



Volatility Persistence and Return Autocorrelation

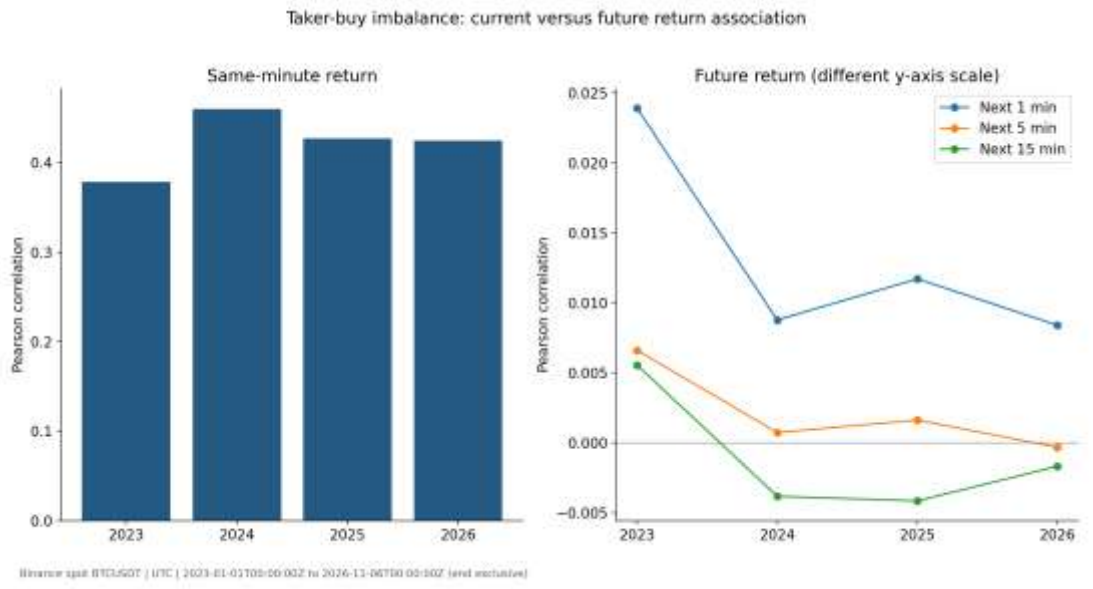


Signed returns retain price direction. Absolute returns measure the size of price changes.

At a one-minute lag, return autocorrelation is approximately zero, while absolute-return autocorrelation is 0.383.

Absolute returns remain positively correlated at longer lags. Intraday seasonality may contribute to this persistence.

Trade Imbalance and Future Returns



$$\text{Imbalance} = 2\times\frac{\text{taker}-\text{buy quote volume}}{\text{total quote volume}} - 1$$

Values near +1 indicate mostly taker buying; values near −1 indicate mostly taker selling.

Pooled correlation is 0.421 with same-minute returns and 0.0125 with next-minute returns.

A strong contemporaneous association still does not by itself establish useful forecasting performance.

Chronological Validation and Statistical Controls

Feature availability: Use only completed-minute features.

Target boundaries: Exclude forecast targets that extend beyond a stage boundary.

Time dependence: Resample consecutive days or weeks in blocks.

Multiple comparisons: Apply Holm adjustment and check results at two block lengths.

*The 2026 period is retrospective because I inspected it before finalising the validation design.

Stage	Period	Purpose
Training	2023–2024	Fit scaling parameters and model coefficients
Validation	2025	Select one candidate per forecast horizon using MSE
Retrospective test	1 Jan–5 Nov 2026	Evaluate selected models with frozen training parameters

Model Choice and Research Purpose

Forecast cumulative returns over the next 1, 5 and 15 minutes using simple models.

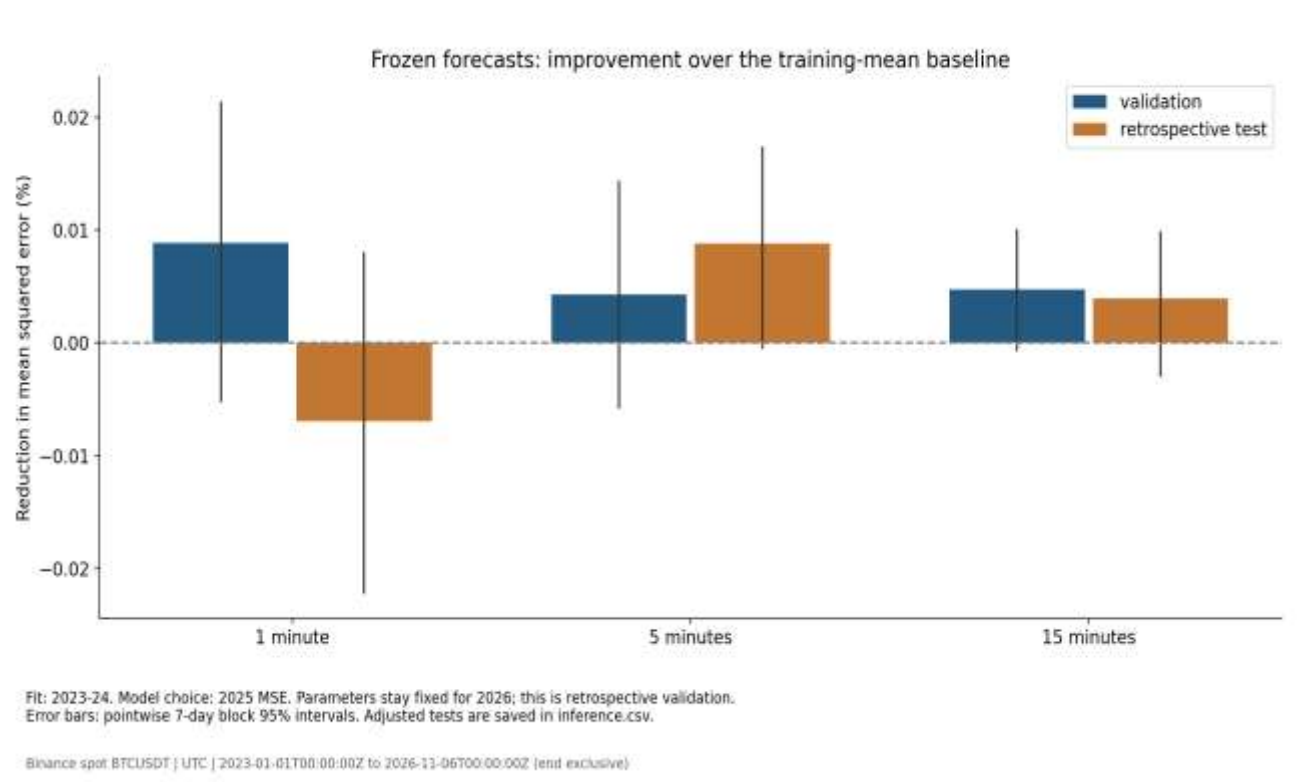
Select the lowest-MSE candidate on 2025 data separately for each horizon.

Evaluate the selected models on 2026 data, keeping the original training parameters fixed.

Candidate	Input	Research purpose
Mean baseline	Training-period mean future return	Benchmark for measuring incremental forecasting value
Return regression	Completed-minute return	Test directional persistence or reversal
Imbalance regression	Completed-minute trade imbalance	Test information about future returns

* Selected using 2025 MSE:
1 minute: Imbalance · 5 minutes: Return · 15 minutes: Return

Evidence Across Chronological Periods



Activity patterns and volatility persistence remain supported across periods.
Robust return-forecasting gains were not established in the 2026 retrospective evaluation.

FINDING	EVIDENCE AFTER VALIDATION
Higher London–New York overlap turnover and variance	Supported in all three periods
One-minute absolute-return persistence	Supported in all three periods
Lower weekend turnover in regional windows	Supported in all three periods
Imbalance and next-minute returns	2026 fails the longer-block Holm check
Forecast MSE improvement	No robust improvement established in 2026

Conclusions and Next Steps

Limitations: Single-exchange kline data limits conclusions about trader locations, order-book conditions and execution costs.

Next Steps: Test whether volume improves future-volatility forecasts beyond past volatility and intraday seasonality.

1. Trading activity and volatility show clear intraday and weekend patterns.

2. Absolute returns show stronger persistence than signed minute returns.

3. Trade imbalance has a clear contemporaneous association with returns, while the tested forecasting models offer limited incremental value.

Thank you for your valuable time;
I look forward to further communication.

Yichen Wang

+44 07756 555022 (UK)

wangyichen20000616@163.com

https://www.linkedin.com/in/yichen-wang-636bb2395/

Appendix - Returns, Volatility and Trade Imbalance

Log return: $r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$

Daily realized variance and volatility: $V_d = \sum_{t \in d} r_t^2$, $RV_d(\%) = 100\sqrt{V_d}$

Trade imbalance:

$$I_m = 2\frac{B_m}{Q_m} - 1$$

Where B_m = taker-buy turnover in minute \(m\), measured in USDT, Q_m = total turnover in the same minute. Notice imbalance is undefined when Q_m = 0.

Zero-volume share: $Z(\%) = 100\frac{\text{observed zero}-\text{volume seconds}}{\text{observed seconds}}$

Appendix - Normalised Activity and Weekend Comparisons

Hourly shares: $s_{d,h}^{X} = 100\frac{X_{d,h}}{X_d}$, $\bar{s}_h^X = \frac{1}{D}\sum_{d=1}^{D} s_{d,h}^{X}$, $X$is BTC volume or realized variance.

Regional-window intensity:

$$I_{d,w}^{X} = \frac{X_{d,w}/M_{d,w}}{X_d/1440}$$

$\boldsymbol{X}$is USDT turnover or realized variance; $M_{d,w}$ is the window duration in minutes.

Weekend / weekday ratio:

$$R_w^X = \frac{\frac{1}{2}\sum_{d\in \mathrm{Sat,Sun}} X_d}{\frac{1}{5}\sum_{d\in \mathrm{Mon,\ldots,Fri}} X_d}$$

Appendix - Forecast Models and Evaluation Metrics

Forecast target:

$$y_{t,h} = \ln\left(\frac{C_{t+h}}{C_t}\right), \qquad h \in \{1,5,15\} \text{ minutes}$$

C_t is the closing price of the completed minute. Each horizon is fitted and evaluated separately.

Historical-mean baseline:

$$\hat{y}_{t,h}^{\text{mean}} = \bar{y}_{h,\text{train}}$$

Single-feature regression:

$$z_t = \frac{x_t - \mu_{x,\text{train}}}{\sigma_{x,\text{train}}}, \qquad \hat{y}_{t,h} = \bar{y}_{h,\text{train}} + \hat{\beta}_h z_t$$

Mean squared error:

$$\text{MSE}_h = \frac{1}{N_h}\sum_t \left(y_{t,h} - \hat{y}_{t,h}\right)^2$$

- Improvement relative to the baseline:

$$\text{Improvement}_h(\%) = 100\left(1 - \frac{\text{MSE}_{h,\text{model}}}{\text{MSE}_{h,\text{baseline}}}\right)$$

Appendix - Block Bootstrap Settings - 1

Statistic	Resampling unit	Primary block	Longer block
Return and imbalance correlations	Calendar day	7 days	28 days
Regional-window intensities	Calendar day	7 days	28 days
Forecast MSE improvement	Calendar day	7 days	28 days
Weekend / weekday ratios	Calendar week	4 weeks	8 weeks

Resampling procedure

Construct a calendar grid within each chronological stage.
Sample consecutive calendar blocks with replacement.
Circular blocks wrap within the same stage.
Recompute the statistic for each resample.

Appendix - Confidence Intervals and Multiple Testing - 1

- Basic bootstrap 95% interval:

$$CI_{95\%} = \left[2\hat{\theta} - q^*_{0.975},\ 2\hat{\theta} - q^*_{0.025}\right]$$

- $\hat{\theta}$ is the observed estimate and q^*_{a} is a quantile of the bootstrap estimates.
- Approximate two-sided bootstrap p-value:

$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\left(\left|\theta^*_b - \hat{\theta}\right| \geq \left|\hat{\theta} - \theta_0\right|\right)}{B + 1}$$

- θ_0 is the null value;B is the number of valid resamples. With 1,999 valid draws, the minimum raw p-value is 0.0005.

Appendix - Confidence Intervals and Multiple Testing - 2

Holm adjustment:

For ordered p-values $p_{(1)} \leq \cdots \leq p_{(m)}$:

$$\tilde{p}_{(i)} = \min\left[1,\ \max_{j \leq i}\{(m-j+1)p_{(j)}\}\right]$$

- Use $m = 39$ within each chronological stage and block-length setting since we have 39 inspections (20 correlations + 8 time-of-day intensity metrics + 8 weekend effect metrics + 3 predictive performance metrics = 39 in total).

Appendix - Correlation Checks: Tests 01–20

- $\rho_X(k) = \mathrm{Corr}(X_t, X_{t+k})\,, \qquad X_t \in \{r_t, |r_t|\}$

ID	Statistic	Lag
01	Return ACF	1 second
02	Return ACF	5 seconds
03	Return ACF	60 seconds
04	Return ACF	300 seconds
05	Absolute-return ACF	1 second
06	Absolute-return ACF	5 seconds
07	Absolute-return ACF	60 seconds
08	Absolute-return ACF	300 seconds

*One-second returns, correlations pool valid observation pairs.

Appendix - Correlation Checks: Tests 01–20 - 2

ID	Statistic	Lag / horizon
09	Return ACF	1 minute
10	Return ACF	5 minutes
11	Return ACF	30 minutes
12	Return ACF	60 minutes
13	Absolute-return ACF	1 minute
14	Absolute-return ACF	5 minutes
15	Absolute-return ACF	30 minutes
16	Absolute-return ACF	60 minutes
17	Imbalance vs. same-minute return	Same minute
18	Imbalance vs. future return	Next 1 minute
19	Imbalance vs. future return	Next 5 minutes
20	Imbalance vs. future return	Next 15 minutes

*One-minute returns and imbalance

Appendix - Window and Weekend Checks: Tests 21–36 - 1

Window intensity: H0: Mean daily intensity = 1

ID	Window	Metric
21	Hong Kong	Turnover intensity
22	Hong Kong	Variance intensity
23	London	Turnover intensity
24	London	Variance intensity
25	New York	Turnover intensity
26	New York	Variance intensity
27	London–New York overlap	Turnover intensity
28	London–New York overlap	Variance intensity

*City windows include their overlaps.
The London–New York overlap is also evaluated as a separate bucket.

Appendix - Window and Weekend Checks: Tests 21–36 - 2

H0: Median weekly ratio = 1.

ID	Scope	Metric
29	Whole day	Daily BTC volume
30	Whole day	Daily RV
31	Hong Kong	Turnover per minute
32	Hong Kong	Variance per minute
33	London	Turnover per minute
34	London	Variance per minute
35	New York	Turnover per minute
36	New York	Variance per minute

*Regional comparisons use local 09:00–17:00 windows.
Each weekly ratio compares weekend and weekday daily averages within the same complete week.

Appendix - Forecast Checks and Validation Scope - 1

Forecast checks: H0: Zero MSE improvement relative to the training-mean baseline.

ID	Forecast horizon	Model selected using 2025
37	1 minute	Imbalance regression
38	5 minutes	Return regression
39	15 minutes	Return regression

*Each check evaluates the selected model against the baseline.

Appendix - Forecast Checks and Validation Scope - 2

Defined test family:

Category	Number of checks
Correlations	20
Window intensities	8
Weekend ratios	8
Selected-model forecast gains	3
Total per stage and block setting	**39**

Three chronological stages × two block settings produce 234 test results.